

Spatial-Temporal Representation Learning with Global-Local Alignment for Next POI Recommendation

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Abstract

Next Point-of-Interest (POI) recommendation aims to predict a user’s next visit based on their check-in history and contextual information. Existing methods often disentangle spatial and temporal factors to model user behaviors, yet they typically overlook two critical issues: 1) the hierarchical structure of mobility trajectories and the regularity in daily routines; 2) the collaboration between global urban dynamics and local user-level patterns. To address these, we propose Spatial-Temporal Representation Learning with Global-Local Alignment (STRL-GLA). Our framework jointly models global spatio-temporal contexts via geographical graph convolution and trajectory-based hypergraph learning, and captures local patterns through user-specific sequential hypergraphs. Moreover, the global-local alignment mechanism enhances knowledge transfer and alleviates interaction sparsity. Extensive experiments on two Location-Based Social Network (LBSN) datasets show that STRL-GLA outperforms state-of-the-art methods.

Code — <https://anonymous.4open.science/r/STRL-GLA>

Introduction

The proliferation of location-based social networks (LBSNs), such as Foursquare, Gowalla, and Yelp, has generated an extensive corpus of user check-in data, reflecting individual activities at Points-of-Interest (POIs) like restaurants, shopping centers, and recreational venues (Zhang et al. 2025). As mobile technology continues to advance, the demand for accurate and context-aware POI recommendations has intensified, necessitating innovative methodologies that transcend traditional approaches.

Early efforts on next POI recommendation relied on sequential modeling techniques, such as Markov chains, to capture the temporal dependencies in user check-in sequences (Cheng et al. 2013). Subsequent advancements incorporated recurrent neural networks (RNNs), long short-term memory (LSTM), and their variants to model long-term and short-term preferences more effectively (Li et al. 2020; Wang et al. 2022a). However, sequential-based methods primarily focus on capturing patterns and regularities within

user check-in sequences, often overlooking global contextual factors like spatial proximity, social influences, or user preferences that may significantly impact the recommendation quality. More recently, graph-based methods have been employed to exploit spatial-temporal relationships among POIs and users (Wang et al. 2022b; Zuo et al. 2024). For example, BiGSL (Wang et al. 2024) adopted a bi-level graph structure learning method to discover the hierarchical structures of POIs and mitigate the noise and sparsity issue of POI information. To model complex high-order dependencies among users and POIs, more and more studies (Yan et al. 2023; Lai et al. 2024) have applied hypergraph-based methods to POI recommendation. This approach effectively addresses data sparsity and enhances recommendation accuracy by leveraging global collaborative signals across multiple users’ trajectories.

Despite these progresses, existing models exhibit some limitations. First, many existing methods prioritize learning associations between POIs or between users and POIs, leading to ignorance of higher-order relationships among daily trajectories and the cyclical pattern of human daily activities. Second, most current studies overlook the integration of global and local perspectives. This narrow focus limits their ability to capture the multifaceted nature of user preferences. In addition, existing methods fail to effectively align representations from global and local views, hindering knowledge transfer and synergy between these views and leading to sub-optimal performance in next POI recommendations.

In this paper, we propose a novel framework Spatial-Temporal Representation Learning with Global-Local Alignment (STRL-GLA) to address above issues. Firstly, we design a local sequential modeling module. Powered by a user sequential hypergraph, it captures daily activity patterns by constructing hyperedges from users’ daily POI visits to learn their dynamic regularities. Secondly, by constructing a geographical graph and a mobility hypergraph, we capture the geographical proximity and high-order trajectory correlations at a global scale. In particular, we propose cross-view POI-level and user-level contrastive learning and the adaptive fusion strategies to optimize global POI and user representations. Then, to form a holistic understanding of users’ global and local patterns, we propose a novel method for aligning users’ global and local preferences, incorporating both distribution alignment and preference discrimination.

Finally, we adaptively fuse user representations from above three graph learning modules for next POI recommendation.

In summary, the main contributions of this paper are as follows:

- We introduce a multi-view POI recommendation method named STRL-GLA to capture both the global trend and the local patterns of human mobility.
- We propose a disentangled graph learning module to learn geographical correlations and inter-sequence interactions. Specially, we adopt a bi-level contrastive learning strategy with adaptive fusion to optimize global user and POI representations.
- We introduce a local sequential hypergraph learning module to discover intra-sequence interactions and daily activity regularity. In addition, we design a global-local alignment method to capture complementary signals between both views.
- Extensive experiments on two major public datasets have demonstrated the superiority of STRL-GLA compared to various state-of-art baselines for next POI recommendation.

Related Work

In early research on POI recommendation, some works rely on Markov chains (Cheng et al. 2013) to model the transition probability to predict the next POI. Other works adopt classic recommendation algorithm such as collaborative filtering (Zheng et al. 2010) and matrix factorization (Mnih and Salakhutdinov 2007). Collaborative filtering leverages user-POI interaction records to identify similar users or locations for generating personalized recommendations, while matrix factorization further enhances this by decomposing the interaction matrix into latent factors to capture underlying patterns, thereby improving recommendation accuracy and addressing sparsity issues.

In recent years, researchers have focused on deep learning methods such as RNN, LSTM, and transformer to capture sequential patterns of POI sequences. For instance, DeepMove (Feng et al. 2018) employs attentional recurrent networks to model multi-level periodicity and long-range dependencies in user mobility trajectories. Similarly, STGN (Zhao et al. 2022) integrates spatio-temporal gating mechanisms into LSTM to effectively capture long and short-term preferences in check-in sequences. Additionally, transformer-based architectures have gained prominence for their ability to handle complex sequential patterns through self-attention mechanism. GETNext (Yang, Liu, and Zhao 2022) enhances the transformer with trajectory flow maps to improve next POI predictions, while AutoMTN (Qin et al. 2022) incorporates auto-correlation and multi-modal features into transformer for more accurate recommendations. More recently, ROTAN (Feng et al. 2024) introduces rotation-based temporal attention to address time-specific aspects in POI recommendation.

Furthermore, Graph Neural Networks (GNNs) have been widely adopted to model the intricate relational structures in POI data, including user-POI interactions and spatio-temporal dependencies. STGCN (Han et al. 2020) con-

structs multi-graphs to represent contextual factors and applies graph convolutions to learn time-dependent representations. ASGNN (Wang et al. 2021) proposes a personalized hierarchical attention mechanism within GNNs to capture varying levels of user-POI correlations. Graph-Flashback (Rao et al. 2022) builds a spatio-temporal knowledge graph, generates weighted POI transition graphs, and applies a lightweight Graph Convolutional Network (GCN) to enrich POI embeddings with historical and contextual signals for improved next-location prediction. Moreover, some recent works have begun exploring hypergraphs for POI recommendation to capture higher-order relationships beyond pairwise interactions. STHGCN (Yan et al. 2023) employs a hypergraph transformer to handle sequential POI data, improving performance in sparse scenarios. ASTHL (Lai et al. 2024) adopts disentangled hypergraph contrastive learning to capture users’ multi-view preferences. However, most of above hypergraph-based methods model user behavior at a global view. They have not implemented a hypergraph on the individual trajectory to capture users’ dynamic preferences at a daily level.

Problem Formulation

Let $\mathcal{U} = \{u_1, u_2, \dots, u_{|\mathcal{U}|}\}$ denote the set of users and $\mathcal{V} = \{v_1, v_2, \dots, v_{|\mathcal{V}|}\}$ represent the collection of Points-of-Interest (POIs), where each POI $v \in \mathcal{V}$ is associated with geographical coordinates (longitude, latitude). For any user $u \in \mathcal{U}$, their historical check-in sequence is organized into sessions based on a time interval (e.g., one day), forming a trajectory sequence $S^u = \{S_1^u, S_2^u, \dots, S_n^u\}$, where each session $S_j^u = \{(l_j, t_j) \mid j = 1, 2, \dots\}$ consists of POI-time pairs indicating visits to l_j at timestamp t_j . Given the above historical data, the goal of next POI recommendation problem is to predict the next POI from $\mathcal{V}$ that user u is the most likely to visit.

Methodology

In this section, we provide a detailed description of our proposed framework as shown in Figure 1.

Local User Modeling

We design a local user modeling module to capture personalized mobility patterns. In this module, we model the user’s daily mobility patterns as a local hypergraph $\mathcal{G}_L^u = (\mathcal{V}_L^u, \mathcal{E}_L^u)$, where the node set $\mathcal{V}_L^u$ consists of all unique POIs the user u has visited, and each hyperedge $e \in \mathcal{E}_L^u$ connects the set of POIs visited by user u on a specific day. The incidence matrix $\mathbf{H}_L^u \in \mathbb{R}^{|\mathcal{V}_L^u| \times |\mathcal{E}_L^u|}$ is formally defined by its entries:

$$h(v, e) = \begin{cases} 1, & \text{if POI } v \in e \\ 0, & \text{otherwise} \end{cases}$$

This matrix precisely captures the relationships between POI nodes and daily trajectory hyperedges.

To capture the complex, higher-order dependencies within each day’s itinerary, we employ a hypergraph convolutional network (HGCN) (Feng et al. 2019) to learn the local POI

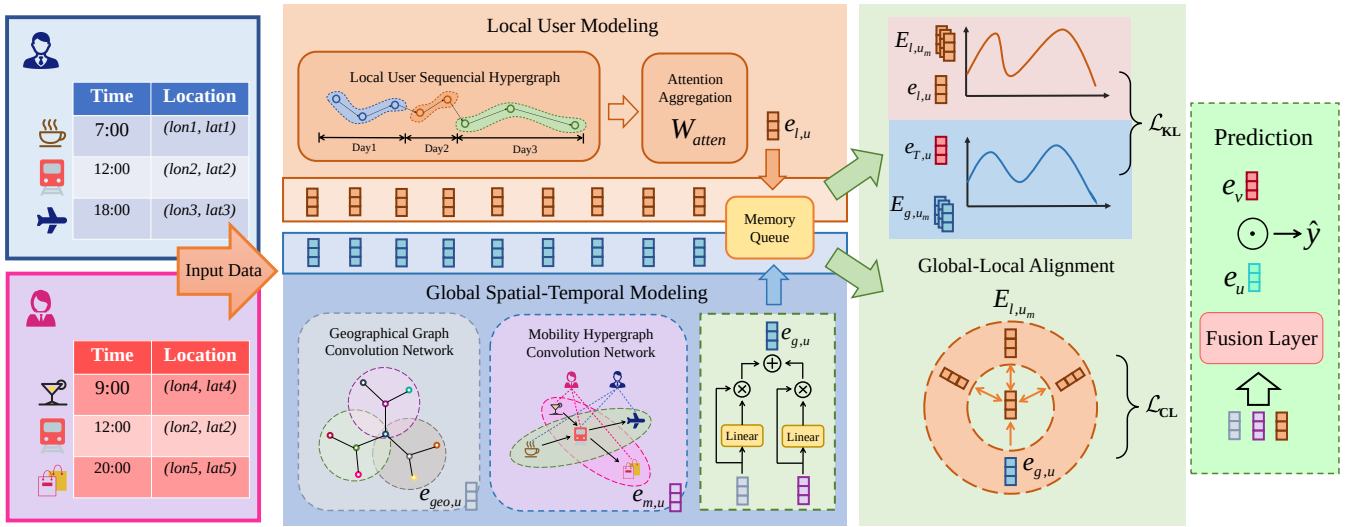


Figure 1: An illustration of the proposed STRL-GLA model.

embeddings. The node-to-node propagation rule in $\mathcal{G}_L^u$ at layer l is formulated as:

$$\mathbf{X}_L^{(l+1)} = \sigma \left(\mathbf{D}_v^{-1/2} \mathbf{H}_L^u \mathbf{W}_L \mathbf{D}_e^{-1} (\mathbf{H}_L^u)^\top \mathbf{D}_v^{-1/2} \mathbf{X}_L^{(l)} \Theta^{(l)} \right)$$

where $\mathbf{X}_L^{(l)}$ is the node embedding matrix at layer l , $\mathbf{D}_v$ and $\mathbf{D}_e$ are the diagonal degree matrices of nodes and hyperedges respectively, $\mathbf{W}_L$ is the hyperedge weight matrix, and $\Theta^{(l)}$ is a learnable parameter matrix.

After propagating through L layers, we obtain a series of node representations $\{\mathbf{X}_L^{(1)}, \mathbf{X}_L^{(2)}, \dots, \mathbf{X}_L^{(L)}\}$. The final embedding for each POI node is computed as the mean of its embeddings across all layers:

$$\mathbf{x}_{L,i} = \frac{1}{L+1} \sum_{l=0}^L \mathbf{x}_{L,i}^{(l)}$$

Subsequently, we aggregate the final learned local POI embeddings from the user's entire historical sequence using an attention mechanism. The attention weight α_i for the i -th POI visit is computed as:

$$\alpha_i = \frac{\exp(\mathbf{q}^\top \cdot \tanh(\mathbf{W}_a \mathbf{x}_{L,i} + \mathbf{b}_a))}{\sum_{j=1}^{|\mathcal{V}_L^u|} \exp(\mathbf{q}^\top \cdot \tanh(\mathbf{W}_a \mathbf{x}_{L,j} + \mathbf{b}_a))}$$

where $\mathbf{W}_a, \mathbf{b}_a, \mathbf{q}$ are learnable parameters. The final local user embedding $e_{l,u}$ is the weighted sum:

$$e_{l,u} = \sum_{i=1}^{|\mathcal{V}_L^u|} \alpha_i \mathbf{x}_{L,i}$$

In this way, we adaptively fuse the user's intrinsic mobility preference at a fine-grained, sequential level.

Global Spatial-Temporal Modeling

To capture the collective mobility patterns and high-order correlations across different users' trajectories, we design a global spatial-temporal modeling module from two complementary perspectives: geographical proximity and user mobility hypergraph.

Geographical Graph We construct a geographical graph $\mathcal{G}_G = (\mathcal{V}_G, \mathcal{E}_G)$ where the node set $\mathcal{V}_G$ includes all POIs in the POI set $\mathcal{V}$ and the edge $e \in \mathcal{E}_G$ connects any POI pairs within a specific distance threshold δ_d .

The message propagation mechanism with residual connections in the geographical graph is defined as:

$$\mathbf{x}_{G,v}^{(l+1)} = \sigma \left(\sum_{j \in \mathcal{N}(v)} \frac{w(d_{vj})}{\sqrt{d_v d_j}} \mathbf{W}_G^{(l)} \mathbf{x}_{G,j}^{(l)} \right) + \mathbf{x}_{G,v}^{(l)}$$

where $\mathcal{N}(v)$ denotes the neighbors of POI v , d_v is the degree of node v , $w(d_{vj}) = e^{-\gamma d_{vj}^2}$ represents the influence weight that decreases exponentially with increasing distance d_{vj} (γ is a predefined hyperparameter), and $\mathbf{W}_G^{(l)}$ is a learnable parameter matrix at layer l .

After propagating through L layers, we obtain a series of node representations $\{\mathbf{x}_{G,v}^{(1)}, \mathbf{x}_{G,v}^{(2)}, \dots, \mathbf{x}_{G,v}^{(L)}\}$. Similar to the local hypergraph module, the final geographical POI embedding for each POI v is computed as the mean of its embeddings across all layers:

$$\mathbf{e}_{geo,v} = \frac{1}{L+1} \sum_{l=0}^L \mathbf{x}_{G,v}^{(l)}$$

This module learns POI representations that capture global geographical relationships.

Mobility Hypergraph We build a mobility hypergraph $\mathcal{G}_M = (\mathcal{V}_M, \mathcal{E}_M)$ where the node set $\mathcal{V}_M$ contains all POIs, and each hyperedge $e \in \mathcal{E}_M$ connects the set of POIs visited by a user in a complete trajectory. The incidence matrix $\mathbf{H}_M \in R^{|\mathcal{V}_M| \times |\mathcal{E}_M|}$ is defined as:

$$h(v, e) = \begin{cases} 1, & \text{if POI } v \in e \\ 0, & \text{otherwise} \end{cases}$$

Similar to the node-to-node propagation rule we leverage in the local user modeling module, we design the mobility

hypergraph convolutional network as follows:

$$\mathbf{X}_M^{(l+1)} = \sigma \left(\mathbf{D}_{v,M}^{-1/2} \mathbf{H}_M \mathbf{W}_M \mathbf{D}_{e,M}^{-1} \mathbf{H}_M^\top \mathbf{D}_{v,M}^{-1/2} \mathbf{X}_M^{(l)} \Theta_M^{(l)} \right)$$

where $\mathbf{D}_{v,M}$ and $\mathbf{D}_{e,M}$ are the diagonal degree matrices of nodes and hyperedges respectively, $\mathbf{W}_M$ is the hyperedge weight matrix, and $\Theta_M^{(l)}$ is a learnable parameter matrix.

After L layers of propagation, we obtain the complex correlations between users' trajectories. Similar to the local hypergraph module, we average the embeddings generated from each layer to obtain the final POI representation $\mathbf{e}_{m,v}$.

Gated Fusion with Contrastive Learning For each user $u \in \mathcal{U}$, the global user embeddings from both views are obtained by summing the corresponding POI embeddings along their historical trajectory. We then fuse the two user representations using a gating mechanism:

$$\mathbf{e}_{g,u} = \lambda_{geo} \cdot \mathbf{e}_{geo,u} + \lambda_m \cdot \mathbf{e}_{m,u}$$

where λ_{geo} and λ_m are learnable gating coefficients. In addition, for each POI $v \in \mathcal{V}$, we sum up the POI embeddings from two decoupled views to obtain the final POI embedding as $e_v = e_{geo,v} + e_{m,v}$.

To enhance consistency between the two global views, we apply contrastive learning at both the POI and user levels. For the same POI/user, the representations from the geographical graph and mobility hypergraph serve as positive pairs, while different POIs/users form negative pairs. This alignment promotes knowledge exchange and collaborations between the two global views.

$$\mathcal{L}_{CL}^P = \frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} -\log \frac{\exp(s(e_{geo,v}, e_{m,v})/\tau)}{\sum_{v' \in \mathcal{V}} \exp(s(e_{geo,v}, e_{m,v'})/\tau)}$$

$$\mathcal{L}_{CL}^U = \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} -\log \frac{\exp(s(e_{geo,u}, e_{m,u})/\tau)}{\sum_{u' \in \mathcal{U}} \exp(s(e_{geo,u}, e_{m,u'})/\tau)}$$

where $\mathcal{L}_{CL}^P$ is the contrastive loss at POI level while $\mathcal{L}_{CL}^U$ is the contrastive loss at the user level, $s(\cdot, \cdot)$ denotes the cosine similarity and τ denotes the temperature of contrastive learning.

Global-Local Alignment

In this section, we present the global-local alignment module to bridge the representation gap between global collective patterns and local sequential behaviors. We implement two complementary alignment strategies to achieve this goal: 1) Distribution alignment operates at a structural level to enforce consistency between global and local views; 2) Preference discrimination performs a direct contrastive adjustment at the instance level.

A key challenge here is to achieve effective alignment without direct pairwise similarity computations between all users, which is computationally prohibitive. To address this, we introduce a memory queue $\mathcal{M}$ that maintains a dynamic set of anchor samples, storing both global and local user embeddings from recent training batches. This queue serves two primary purposes: 1) it provides a rich and efficient sampling of the broader population's distribution for contextual comparison, and 2) it enables stable and consistent alignment by leveraging a large number of negative samples across batches.

Distribution Alignment To align the semantic distributions between the global and local views, we adopt a distribution-level alignment strategy. Specifically, we maintain a memory queue $\mathcal{M}$ that stores the global user embeddings $\mathbf{e}_{g,u_m}$ and local user embeddings $\mathbf{e}_{l,u_m}$ of M randomly sampled users $\{u_1, u_2, \dots, u_M\}$, which serve as anchor samples.

For a given user u , we first obtain the embedding of the next POI of the user, denoted as $\mathbf{e}_{T,u}$, from the fused global POI representations. We then compute the similarity distribution between $\mathbf{e}_{T,u}$ and all global anchor user embeddings $\mathbf{e}_{g,u_m}$ in the memory queue $\mathcal{M}$. Similarly, we compute the similarity distribution between the user's local embedding $\mathbf{e}_{l,u}$ and all local anchor user embeddings $\mathbf{e}_{l,u_m}$ in $\mathcal{M}$. The similarity distributions are formulated as:

$$p_l^m = \frac{s(\mathbf{e}_{l,u}, \mathbf{e}_{l,u_m})}{\sum_{m'=1}^M s(\mathbf{e}_{l,u}, \mathbf{e}_{l,u_{m'}})}$$

$$p_g^m = \frac{s(\mathbf{e}_{T,u}, \mathbf{e}_{g,u_m})}{\sum_{m'=1}^M s(\mathbf{e}_{T,u}, \mathbf{e}_{g,u_{m'}})}$$

The objective of distribution alignment is to enforce consistency between local preference distribution and the global contextual distribution of the next POI. We achieve this by minimizing the Kullback-Leibler divergence between the two distributions:

$$\mathcal{L}_{KL} = \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} D_{KL}(p_g \| p_l)$$

This loss function maximizes the harmonious patterns between local and global views by minimizing the difference between two derived distributions.

Preference Discrimination In this subsection, we apply a direct instance-level contrastive loss to pull the global and local representations of the same user closer while pushing apart representations of different users.

For a user u , we take the global user embedding $\mathbf{e}_{g,u}$ as the positive sample of the local user embedding $\mathbf{e}_{l,u}$ while the local embeddings $\mathbf{e}_{l,u_m}$ of other users in the memory queue $\mathcal{M}$ are treated as negative samples. The preference discrimination loss is defined as:

$$\mathcal{L}_{CL} = -\log \frac{\exp(s(\mathbf{e}_{l,u}, \mathbf{e}_{g,u})/\tau)}{\sum_{u_m \in \mathcal{M}} \exp(s(\mathbf{e}_{l,u}, \mathbf{e}_{l,u_m})/\tau)}$$

In the end, the total loss for the global-local alignment module is a weighted combination of the distribution alignment loss and the preference discrimination loss:

$$\mathcal{L}_{Align} = \mathcal{L}_{KL} + \alpha \cdot \mathcal{L}_{CL}$$

where α is a hyperparameter balancing the two objectives. This combined loss ensures alignment at both the distributional and instance levels.

Prediction and Joint Optimization

Similar to gated fusion module, the final user representation $\mathbf{e}_u$ is obtained by fusing the global user embedding $\mathbf{e}_{g,u}$ and the local user embedding $\mathbf{e}_{l,u}$ using a gating mechanism:

Method	NYC				TKY			
	Recall@5	Recall@10	NDCG@5	NDCG@10	Recall@5	Recall@10	NDCG@5	NDCG@10
STGN	0.2371	0.2594	0.2261	0.2307	0.2112	0.2587	0.1482	0.1589
LSTPM	0.2495	0.2668	0.2425	0.2483	0.2203	0.2703	0.1556	0.1734
STAN	0.3523	0.3827	0.3025	0.3137	0.2621	0.3317	0.2074	0.2189
LightGCN	0.3221	0.3488	0.2958	0.3042	0.2213	0.2594	0.1977	0.2098
SGRec	0.3451	0.3723	0.3052	0.3178	0.2537	0.3213	0.2221	0.2447
GETNext	0.3572	0.3866	0.3079	0.3094	0.2686	0.3282	0.2212	0.2242
DHCN	0.3745	0.3966	0.3126	0.3203	<u>0.3172</u>	0.3454	<u>0.2442</u>	<u>0.2543</u>
DisenPOI	0.3577	0.3831	0.2979	0.3071	<u>0.2692</u>	0.3314	<u>0.2263</u>	<u>0.2332</u>
ASTHL	0.4119	0.4477	<u>0.3597</u>	<u>0.3707</u>	0.2967	0.3509	0.2358	0.2506
STRL-GLA	0.4395	0.4733	0.3886	0.3952	0.3358	0.3908	0.2590	0.2777
%Improv	+6.70	+5.72	+8.03	+6.61	+5.86	+11.37	+6.06	+9.20

Table 1: Performance comparison on two datasets in terms of Recall and NDCG. The best and the second-best performances are bolded and underlined, respectively. The relative improvements are calculated between the best and the second-best scores.

$$\mathbf{e}_u = \lambda_{geo} \cdot \mathbf{e}_{geo,u} + \lambda_m \cdot \mathbf{e}_{m,u} + \lambda_l \cdot \mathbf{e}_{l,u}$$

where λ_{geo} and λ_m are learnable gating coefficients from the gated fusion module and λ_l is the gating coefficient that can be learned for the embedding of the local user. The preference score of user u for a candidate POI v is calculated via the dot product of their final representations as $\hat{y}_{u,v} = \mathbf{e}_u^\top \mathbf{e}_v$. Finally, we formulate the learning objective as a cross-entropy loss function:

$$\mathcal{L}_{Rec} = - \sum_{u \in \mathcal{U}} \sum_{v \in \mathcal{V}} y_{u,v} \log(\hat{y}_{u,v}) + (1 - y_{u,v}) \log((1 - \hat{y}_{u,v}))$$

The model is trained end-to-end by combining the recommendation loss with the contrastive and alignment losses. The primary recommendation loss $\mathcal{L}_{Rec}$ is the cross-entropy loss between the predicted scores and the ground-truth next POI. The final training objective is:

$$\mathcal{L} = \mathcal{L}_{Rec} + \lambda_{CL}(\mathcal{L}_{CL}^P + \mathcal{L}_{CL}^U) + \lambda_A \mathcal{L}_{Align} + \lambda_\theta \|\Theta\|_2$$

where λ_{CL} and λ_A are hyperparameters controlling the contribution of each auxiliary loss, $\|\Theta\|_2$ denotes the L2 regularization of all parameters for preventing over-fitting issue under the control of λ_θ .

Experiments

Experimental Settings

Datasets To evaluate the performance of our proposed model, we conduct experiments using two widely adopted real-world datasets collected from Foursquare: the New York City (NYC) and Tokyo (TKY) datasets. These datasets contain check-in records spanning from April 12, 2012, to February 16, 2013. We filter out inactive users with fewer than 10 check-ins and remove unpopular POIs visited by fewer than 10 users. To maintain temporal coherence, trajectories are segmented by a 24-hour interval and sequences shorter than 2 check-ins are discarded.

Metrics To evaluate the top- K recommendation performance, we employ Recall@ K and Normalized Discounted Cumulative Gain (NDCG@ K) with K values of $\{5, 10\}$. These metrics measure the proportion of relevant items captured in the top- K recommendations and assess the ranking quality of the recommendation lists.

Baselines Following baseline models are selected for comparison: 1) sequential-based methods STGN (Zhao et al. 2022), LSTPM (Jiang et al. 2023), and STAN (Luo, Liu, and Liu 2021); 2) graph-based methods LightGCN (He et al. 2020), SGRec (Li et al. 2021) and GETNext (Yang, Liu, and Zhao 2022); multi-view hypergraph learning based methods DHCN (Xia et al. 2021), DisenPOI (Qin et al. 2023) and ASTHL (Lai et al. 2024).

Implementations We implement our model with PyTorch 2.9 and evaluate it on a workstation with an RTX4090 GPU. We adopt an Adam optimizer with a learning rate of 0.001. λ_{CL} is set to 0.1, λ_A is 0.1, α is 0.8, and λ_θ is 0.00001. The embedding dimension d is 128, and the batch size is 200. We set the distance threshold δ_d as 2.5km for NYC and TKY. The number of layers L is set to 3, and the temperature for contrastive learning and preference discrimination is 0.1.

Results and Analysis

The comprehensive experimental results is presented in Table 1. It reveals the superior performance of our proposed STRL-GLA framework across two real-world datasets. We attribute the improvements to the following factors: 1) By disentangling global and local views, our model can capture the holistic preference of each individual user. 2) Hypergraph convolutional network can alleviate data sparsity more effectively than graph neural network and sequential-based methods. 3) Benefiting from our proposed global-local alignment method, local modeling module can distill complementary recommendation signals from the global view. Moreover, we observe that all methods perform better on NYC dataset than TKY dataset since TKY dataset have more POIs and a larger user population. Therefore, TKY dataset presents a challenge with more complex user-POI interactions. The results show that our model demonstrates a

Method	NYC		TKY	
	Recall@10	NDCG@10	Recall@10	NDCG@10
w/o G	0.4545	0.3316	0.3829	0.2486
w/o M	0.4418	0.3685	0.3581	0.2614
w/o L	0.4574	0.3784	0.3619	0.2541
w/o C	0.4655	0.3896	0.3687	0.2637
w/o A	0.4574	0.3899	0.3587	0.2599
STRL-GLA	0.4733	0.3952	0.3908	0.2777

Table 2: Ablation study on STRL-GLA.

more remarkable improvement over the previous state-of-art method on TKY dataset. It once again demonstrates that our model performs exceptionally on capturing intricate user-POI correlations and complementary recommendation signals.

Ablation Studies

In this section, we conduct experiments to evaluate component-specific benefits from following: 1) **w/o G**, which removes the geographical graph convolution network; 2) **w/o M**, which removes the mobility hypergraph convolution network; 3) **w/o L**, which removes local user modeling module; 4) **w/o C**, which removes user and POI level contrastive learning; 5) **w/o A**, which removes the global-local alignment. The results are shown in Table 2. It shows that after removing the geographical graph convolution network or the mobility hypergraph convolution network, the model performance drops clearly. Therefore, each component of the global view can capture comprehensive user-POI interactions. Second, after removing the local user modeling module, performances drop slightly compared to other variants. It proves that global user-POI interactive signals can be vital for individual preference modeling. Additionally, the removal of contrastive loss has the least impact on the model’s performance, but it leads to suboptimal recommendation effects. Finally, the removal of global-local alignment leads to similar degradation in performances as the removal of the local user modeling module. It indicates that global-local alignment can distill complementary information from the global view to enhance personalized interest modeling.

Moreover, we conduct experiments to verify the effectiveness of distribution alignment and preference discrimination in global-local alignment. As illustrated in Figure 2, removing either distribution alignment (w/o DA) or preference discrimination (w/o PD) causes a decline in model’s performance. It indicates that both of them play complementary roles in fusing global and local recommendation signals.

Hyperparameter Sensitivity Analysis

To investigate the impact of different parameter settings on recommendation effectiveness, we discuss two critical hyperparameters in the model. We search for the number of hypergraph layers L within the range $\{1, 2, 3, 4, 5\}$ and λ_A , which controls the strength of global-local alignment, within the range $\{0, 0.001, 0.01, 0.1, 1, 2\}$. To achieve the optimal performance, we evaluate all combinations on two datasets and the final results are presented in Figure 3. Finally we set L to 3 and λ_A to 0.1 in this work.

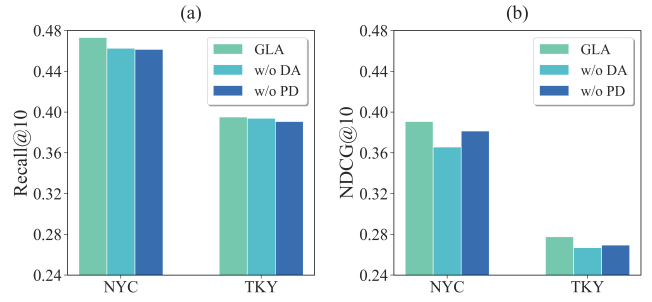


Figure 2: Ablation study on global-local alignment

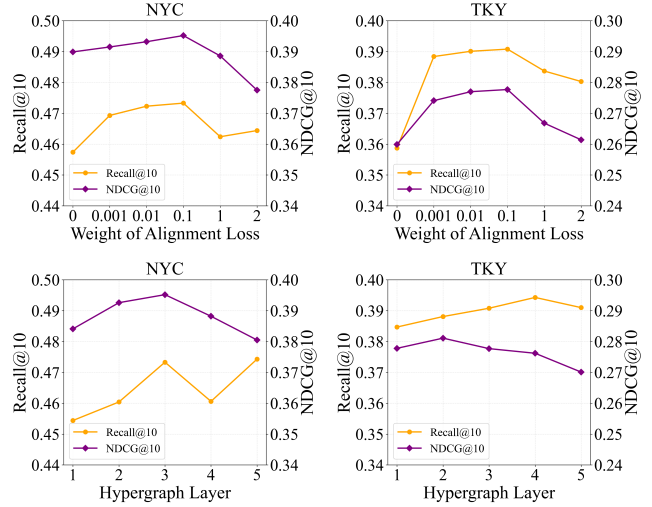


Figure 3: Hyperparameter sensitivity analysis

Conclusion

This paper introduces STRL-GLA for next POI recommendation. Firstly, our model introduces hypergraph learning to capture fine-grained individual preferences from a local view. Secondly, we decouple geographical influences and inter-sequence correlations by constructing a geographical graph and a mobility hypergraph. Additionally, we adopt a bi-level contrastive learning with the adaptive fusion method to refine user representations and a global-local alignment module to model mutual intentions between global and local views. Experimental results on two major datasets demonstrate effectiveness of our framework. For future work, we will explore more potential factors behind complex user-POI interactions and the interpretability of POI recommendation system.

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